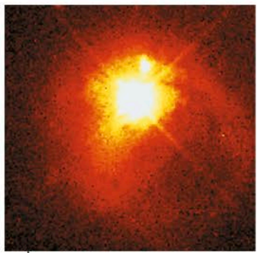
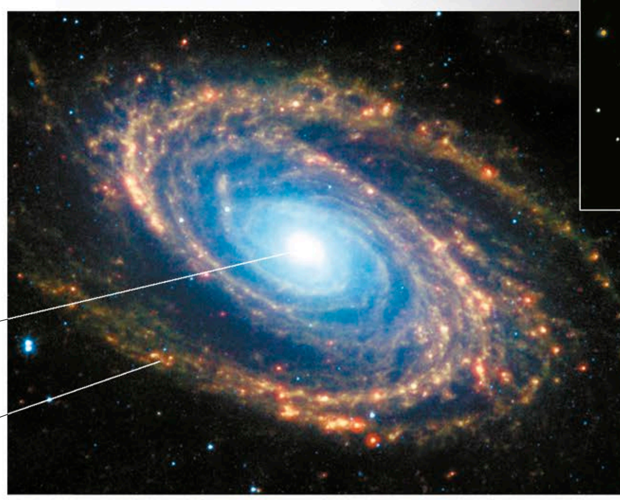
GALAXIES

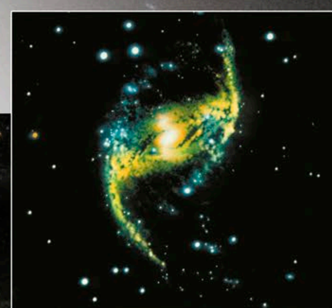


QUASAR
Some, if not most, galaxies are thought to have been quasars earlier in their life. Quasars are extremely luminous galaxies powered by matter falling into a massive, central black hole.

The solar system occupies just a tiny part of an enormous, disk-shaped structure of stars, gas, and dust called the Milky Way Galaxy. Until around a hundred years ago, our galaxy was thought to comprise the whole universe; few people imagined that anything might exist outside of the Milky Way. Today, we know that just the observable part of our universe contains more than 1 trillion separate galaxies. They vary in size from dwarf galaxies, a few hundred light-years across and containing about 100 million stars, to giants spanning several hundred thousand light-years and containing as many as 100 trillion stars. As well as stars, galaxies contain clouds of gas, dust, and dark matter (see opposite), all held together by gravity. They come in five shapes: spiral, barred spiral, elliptical (spherical to football-shaped), lenticular (lens-shaped), and irregular. Astronomers identify galaxies by their number in one of several databases of celestial objects. For example, NGC 1530 indicates galaxy 1530 in a database called the New General Catalog (NGC).

SPIRAL GALAXY
This image taken by the Spitzer Space Telescope shows a nearby spiral galaxy called M81. The sensor captured infrared radiation rather than visible light, and the image highlights dust in the galaxy's nucleus and spiral arms.

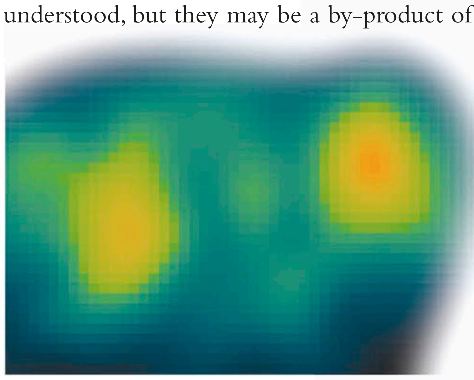




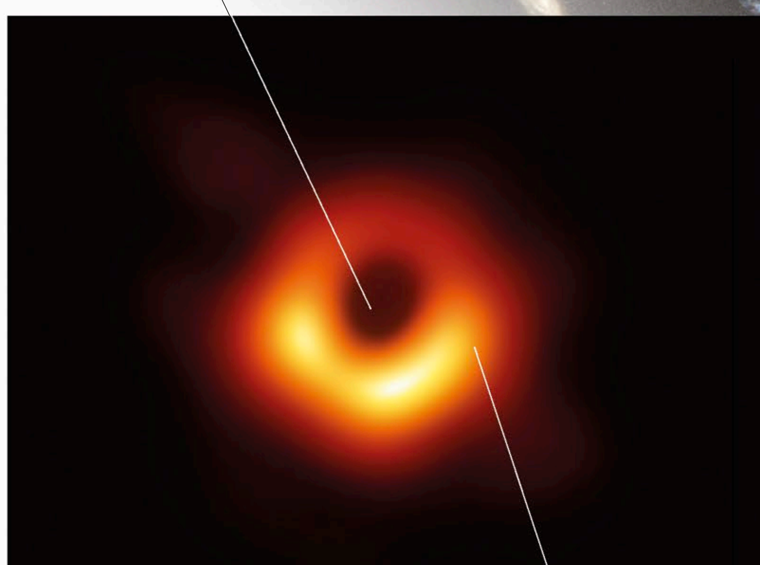
BARRED SPIRAL
In a barred spiral galaxy, such as NGC 1530 (above), the spiral arms radiate from the ends of the central barlike structure rather than from the nucleus.

BLACK HOLES

A black hole is a region of space containing, at its center, some matter squeezed into a point of infinite density, called a singularity. Within a spherical region around the singularity, the gravitational pull is so great that nothing, not even light, can escape. Black holes can therefore be detected only from the behavior of material around them; those discovered so far typically have a disk of gas and dust spinning around the hole, throwing off hot, high-speed jets of material or emitting radiation (such as X-rays) as matter falls into the hole. There are two main types of black holes: supermassive and stellar. Supermassive black holes, which can have a mass equivalent to billions of suns, exist in the centers of most galaxies, including our own. Their exact origin is not yet understood, but they may be a by-product of the process of galaxy formation. Stellar black holes form from the collapsed remains of exploded supergiant stars (see p.267) and may be very common in all galaxies.



STELLAR BLACK HOLE
The black hole SS 433 is situated in the center of this false-color X-ray image. It is detectable because it is sucking in matter from a nearby star and blasting out material and X-ray radiation, visible here as two bright yellow lobes.



GALACTIC BLACK HOLE
This image, captured in 2019 by the Event Horizon Telescope, was the first to show an overt black hole—a supermassive black hole at the center of galaxy M87.

INTRODUCTION